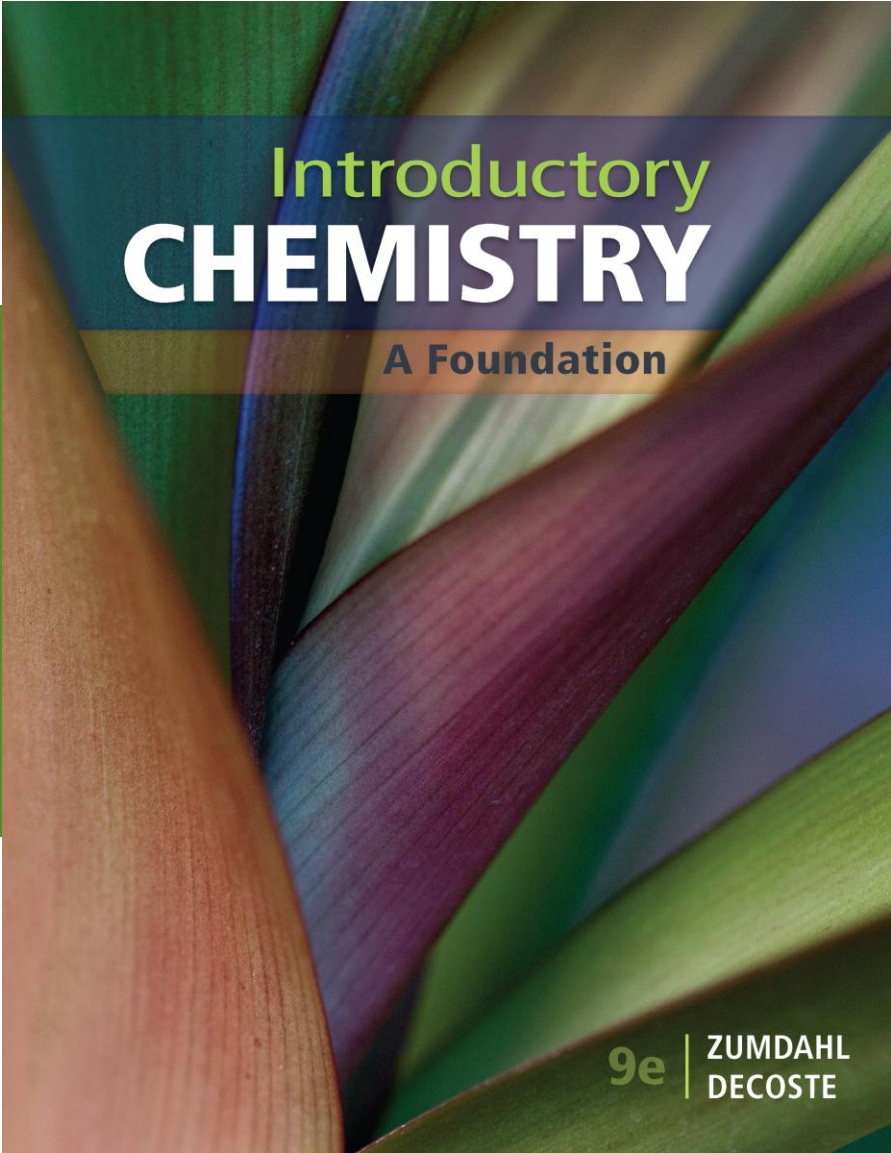




Introductory **CHEMISTRY**

A Foundation

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Chapter 3

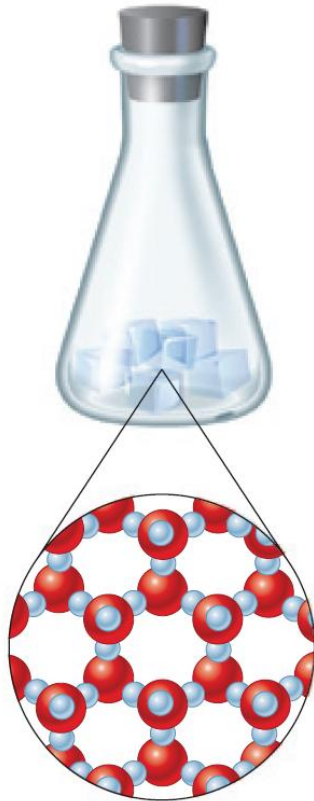
Matter

Matter

Matter

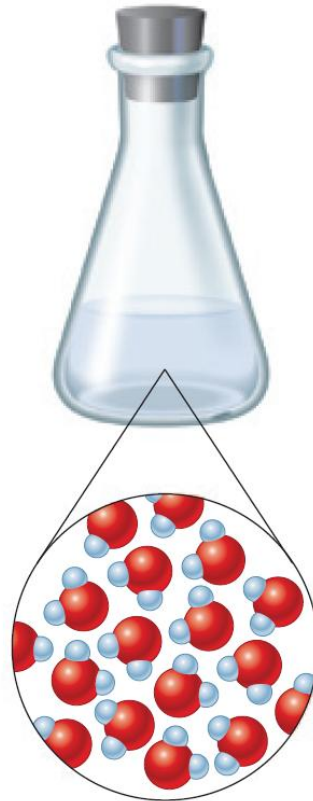
- Anything occupying space and having mass
- Matter exists in three states
 - Solid
 - Liquid
 - Gas

The Three States of Water



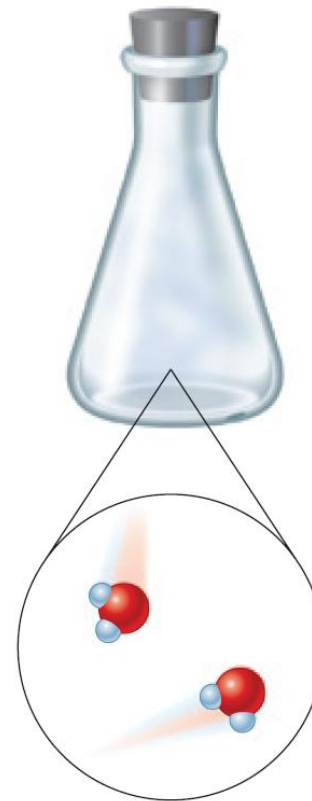
Ice

Solid: The water molecules are locked into rigid positions and are close together.



Water

Liquid: The water molecules are still close together but can move around to some extent.



Steam

Gas: The water molecules are far apart and move randomly.

Matter

Solid

- Rigid
- Has a fixed volume and shape
- Examples
 - Ice cube
 - Diamond
 - Iron bar

Matter

Liquid

- Has a definite volume
- Assumes shape of container
- Examples
 - Gasoline
 - Water
 - Alcohol
 - Blood

Liquid Water Takes the Shape of Its Container

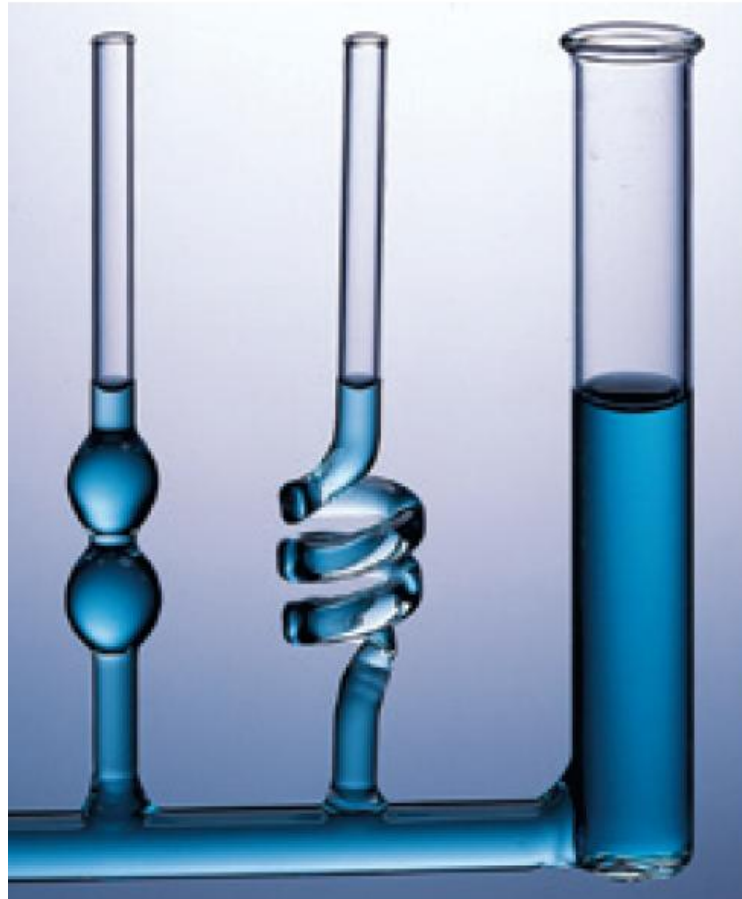


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Matter

Gas

- Has no fixed volume or shape
- Takes the shape and volume of its container
- Examples
 - Air
 - Helium
 - Oxygen

Physical and Chemical Properties and Changes

Physical Properties

- Characteristics that are directly observable and unique to a substance
- Examples
 - Odor
 - Color
 - Volume
 - State
 - Density
 - Melting point
 - Boiling point

Physical and Chemical Properties and Changes

Chemical Properties

- A substance's ability to form new substances
- The characteristics that determine how the composition of matter changes as a result of contact with other matter or the influence of energy
- Characteristics that describe the behavior of matter
- Examples
 - Flammability
 - Rusting of steel
 - The digestion of food

Physical and Chemical Properties and Changes

Concept Check, 1

Classify each of the following as a **physical** or **chemical** property

- Ethyl alcohol boiling at 78°C physical
- Hardness of a diamond physical
- Sugar fermenting to form ethyl alcohol chemical

Physical and Chemical Properties and Changes

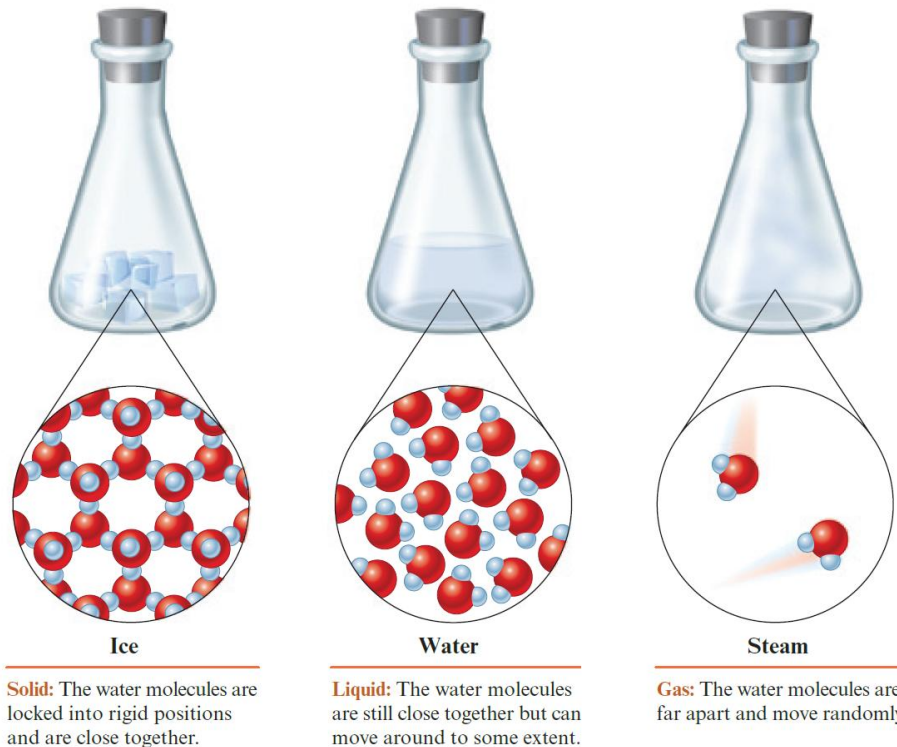
Physical Change

- Change in one or more physical properties of a substance, and not in its chemical composition
- Example: Boiling or freezing water

Physical and Chemical Properties and Changes

Three States of Water

- In all three phases, water molecules are still intact
- Motions of molecules and the distances between them change



Physical and Chemical Properties and Changes

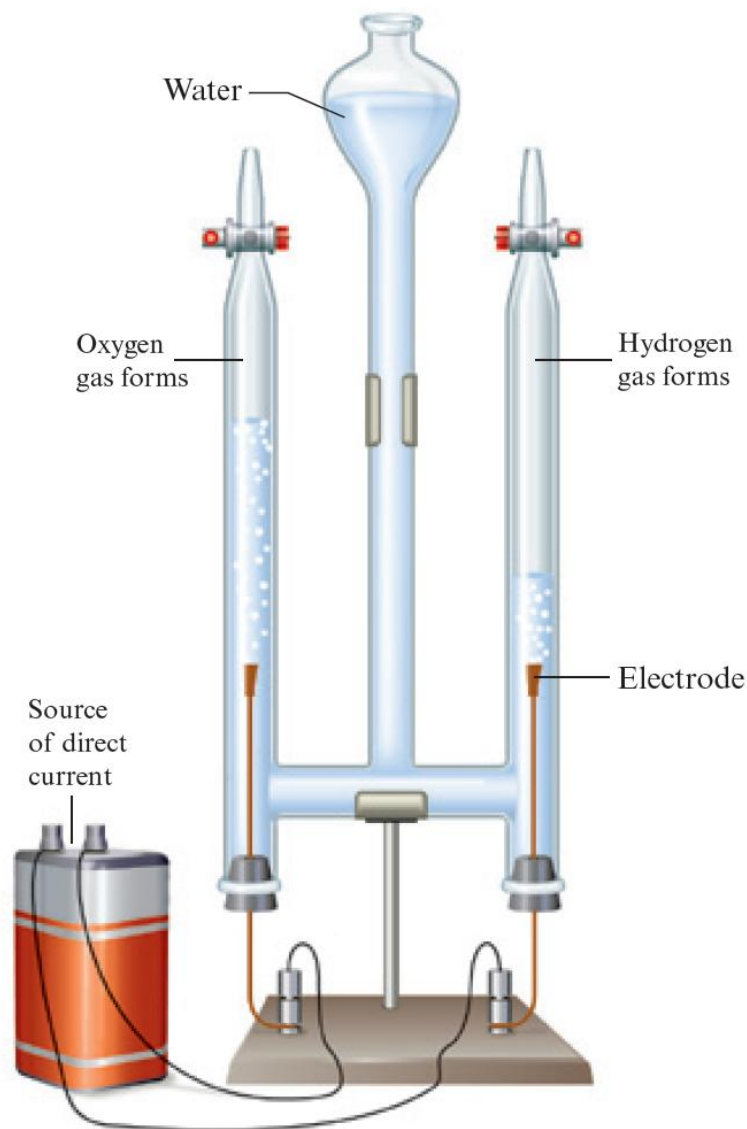
Chemical Change

- A given substance becomes a new substance or substances with different properties and different composition
- Example: Bunsen burner, where methane reacts with oxygen to form carbon dioxide and water

Section 3.2

Physical and Chemical Properties and Changes

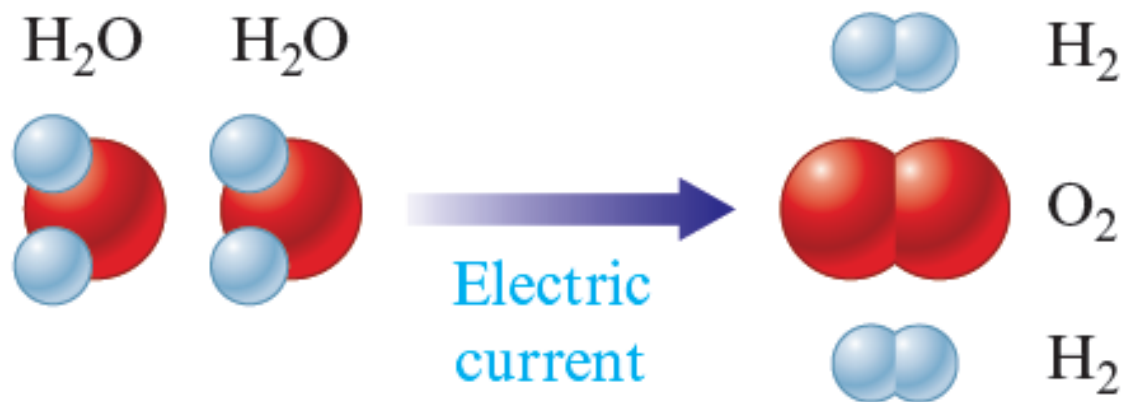
Electrolysis of Water



Physical and Chemical Properties and Changes

Electrolysis of Water (continued)

- Water decomposes to hydrogen and oxygen gases



Physical and Chemical Properties and Changes

Concept Check, 2

Which of the following are examples of a chemical change?

- a) Pulverizing or crushing rock salt
- b) Burning of wood
- c) Dissolving of sugar in water
- d) Melting a popsicle on a warm summer day

Physical and Chemical Properties and Changes

Concept Check, 3

Classify each of the following as a **physical** or **chemical** change

- Sugar fermenting to form ethyl alcohol chemical
- Iron metal melting physical
- Iron combining with oxygen to form rust chemical

Elements and Compounds

Element

- A substance that cannot be broken down into other substances by chemical methods
- Examples
 - Iron, Fe; aluminum, Al; oxygen, O₂; and hydrogen, H₂
- All of the matter in the world around us contains elements

Elements and Compounds

Compound

- A substance composed of a given combination of elements that can be broken down into those elements by chemical methods
- Examples
 - Water, H_2O ; carbon dioxide, CO_2 ; and table sugar, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
- A compound always contains atoms of different elements
- A compound always has the same composition or the same combination of atoms

Elements and Compounds

Concept Check, 4

How many of the following are **compounds**?

H_2O , N_2O_4 , NaOH , MnO_2 , HF

Five - All of the substances are compounds

Mixtures and Pure Substances

Pure Substances

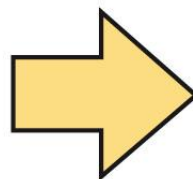
- Always have the same composition
- Are either elements or compounds
- Examples:
 - Pure water or H_2O , carbon dioxide or CO_2 , hydrogen or H_2 , and gold or Au

Mixtures and Pure Substances

Mixtures

- Have variable composition
- Examples
 - Wood, wine, and coffee
- Can be separated into pure substances that are elements and/or compounds

Mixtures



Two or
more pure
substances

Mixtures and Pure Substances

Homogeneous Mixture

- A substance that is the same throughout
- Consists of visibly indistinguishable parts
- Is also called a solution
- Does not vary in composition from one region to another

Mixtures and Pure Substances

Homogeneous Mixture - Examples

- Air around you
- Brass
- Table salt stirred into water

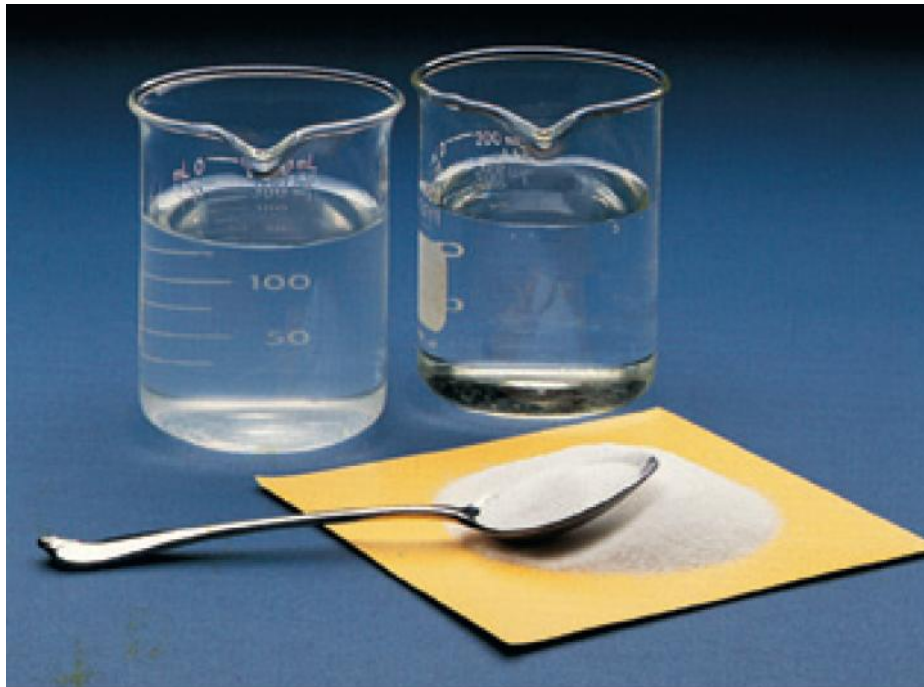


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Mixtures and Pure Substances

Heterogeneous Mixture

- Consists of visibly distinguishable parts
- Contains regions that have different properties from those of other regions

Section 3.4

Mixtures and Pure Substances

Heterogeneous Mixture - Examples

- Oil and vinegar dressing
- Sand stirred into water



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Mixtures and Pure Substances

Concept Check, 5

Which of the following is a **homogeneous mixture**?

- Pure water
- Gasoline
- Jar of jelly beans
- Soil
- Copper metal

Separation of Mixtures

The Organization of Matter

